GENERAL CASE STUDY

PROBLEM STATEMENT

2026

COMPOSIT | 31ST EDITION

SECURING RARE EARTH MATERIALS FOR THE ELECTRIC VEHICLE REVOLUTION

PROBLEM

Electric vehicles (EVs) are central to the global transition toward sustainable transportation. A critical component of many high-performance EV motors is the **permanent magnet synchronous motor (PMSM)**, which relies heavily on **Rare Earth Elements (REEs)** such as **Neodymium (Nd), Praseodymium (Pr),** and **Dysprosium (Dy).**

These rare earth magnets provide extremely high magnetic strength, enabling compact, efficient, and high-power electric motors. However, the global supply chain for rare earth materials is **highly concentrated geographically**, making it vulnerable to geopolitical risks, price volatility, and supply disruptions.

As EV demand grows rapidly, manufacturers face a crucial challenge:
How can they secure a reliable and sustainable supply of rare earth materials while maintaining cost efficiency and environmental responsibility?

BACKGROUND

The global electric vehicle market is projected to grow exponentially in the coming decades, with millions of EVs expected to be produced annually. Most EV traction motors rely on **NdFeB permanent magnets,** which contain rare earth elements such as neodymium and dysprosium.

However, the rare earth supply chain faces several critical challenges:

- **Supply concentration**: A significant portion of global rare earth mining and refining occurs in limited regions, creating dependency risks.
- **Environmental concerns**: Mining and processing of rare earth elements can generate toxic waste and environmental degradation.
- **Price volatility**: Rare earth prices have historically experienced sharp fluctuations due to geopolitical tensions and export restrictions.
- **Recycling limitations**: Current recycling infrastructure for rare earth magnets remains underdeveloped.

As EV production increases, ensuring a **stable**, **ethical**, and **economically viable** rare earth supply chain becomes a strategic priority for automakers and governments worldwide.

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INDUSTRY CONTEXT

Rare earth permanent magnets are widely used in advanced technologies such as electric vehicle motors, wind turbines, consumer electronics, and defense systems.

In EVs, these magnets enable **high torque density**, **improved efficiency**, and **compact motor design**. However, the global rare earth supply chain is highly concentrated, making it vulnerable to geopolitical risks and price fluctuations.

As EV adoption accelerates, companies must explore strategies such as supply diversification, recycling, and material innovation to ensure a secure and sustainable rare earth ecosystem.

DETAILED PROBLEM STATEMENT

Imagine you are part of the strategic innovation team of a leading electric vehicle manufacturer that plans to scale production significantly over the next decade. The company currently relies on **rare earth permanent magnets for its EV motors**, but rising demand and supply chain uncertainties threaten the long-term sustainability of this approach.

Your challenge is to develop a comprehensive strategy to mitigate rare earth supply risks while maintaining **performance**, **cost efficiency**, and **sustainability**.

Your solution should address the following key aspects:

- Rare Earth supply chain overview.
- Risk identification and problem analysis.
- Short-term mitigation strategies that could be implemented in the near future to ensure a stable supply.
- Mid-term technological solutions that could reduce reliance on newly mined rare earth materials.
- Long-term innovation pathways that could fundamentally reduce dependency on rare earth materials and reshape the EV industry over the next two decades.
- Sustainability and environmental impact of rare earth mining and processing.
- Economic feasibility of the proposed strategies. Explain how your strategy balances innovation, cost efficiency and supply security.

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BONUS CHALLENGE

Looking ahead to the future of sustainable mobility -

How might **AI-driven supply chain management, advanced materials science,** and **global recycling networks** transform the rare earth ecosystem for electric vehicles?

Discuss how your company could position itself as a leader in sustainable rare earth innovation.

SUBMISSION RULES

- **Slide Limit** - 10 slides
- **Submission Format** - .pdf only
- **Submission Link** - *General Case Study | COMPOSIT 31st Edition*
- First slide of the presentation must include -
  - Team Name
  - Team ID
  - Members' Names
  - Members' COMPOSIT IDs

EVALUATION CRITERIA

Submissions will be evaluated based on:
- Problem Understanding & Research Depth
- Innovation and Creativity
- Feasibility and Practicality
- Economic and Sustainability Analysis
- Clarity of Presentation

CONTACT

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